

The Root System

Tap Root System

Direct elongation of the embryonic **radicle** forms the primary root, which branches into secondary, tertiary roots. Together they form the **tap root system** (e.g. Mustard).

Fibrous Root System

In monocots the primary root is short-lived and replaced by a dense cluster of roots arising from the stem base, forming the **fibrous root system** (e.g. Wheat).

Adventitious Roots

Roots arising from parts other than the radicle (grass, Monstera, Banyan) are **adventitious roots**.

Functions of Roots

Absorption of water & minerals, anchorage, food storage, and synthesis of growth hormones.

Regions of the Root Tip

Root cap (protective) → **Region of meristematic activity** (small, dividing cells) → **Region of elongation** → **Region of maturation** (root hairs develop here).

Modifications of Roots

Roots can be modified for special functions: **storage roots** (carrot, radish, turnip) swell with reserve food; **prop roots** in banyan provide mechanical support to spreading branches; **pneumatophores** in mangroves aid respiration in waterlogged soil.

Root Nodules and Symbiosis

In leguminous plants, roots form **nodules** housing nitrogen-fixing bacteria (Rhizobium), establishing a symbiotic relationship that enriches soil nitrogen content.

Comparative Importance

While shoot systems capture light and CO₂, root systems anchor the plant and mine the soil for water and minerals — the two systems function as complementary halves of the whole plant body.

Tap Root vs Fibrous Root

Feature	Tap Root	Fibrous Root
Origin	Radicle	Stem base
Plant group	Dicots	Monocots
Example	Mustard	Wheat
Main root	Present, persistent	Absent/short-lived

4 Regions of the Root Tip

- Root cap
- Region of meristematic activity
- Region of elongation
- Region of maturation (root hairs)